Fairness-Aware Multiobjective Evolutionary Learning

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***Abstract*—Multiobjective evolutionary learning (MOEL) has demonstrated its advantages of training fairer machine learning models considering a predefined set of conflicting objectives, including accuracy and different fairness measures. Recent works propose to construct a representative subset of fairness measures as optimization objectives of MOEL throughout model training. However, the determination of a representative measure set relies on the dataset, prior knowledge, and requires substantial computational costs. What is more, those representative measures may differ across different model training processes. Instead of using a static predefined set determined before model training, this article proposes to dynamically and adaptively determine a representative measure set online during the model training. The dynamically determined representative set is then used as optimizing objectives of the MOEL framework and can vary with time. Extensive experimental results on 12 well-known benchmark datasets demonstrate that our proposed framework achieves outstanding performance compared to the state-of-the-art approaches for mitigating unfairness in terms of accuracy as well as 25 fairness measures although only a few of them were dynamically selected and used as optimization objectives. The results indicate the importance of setting optimization objectives dynamically during training.**

***Index Terms*—Artificial neural networks (ANNs), evolutionary computation, fair machine learning, fairness measures, multiobjective learning.**

I. Introduction

FAIRNESS is a critical concern in artificial intelligence [1], [2], [3], [4]. Over the years, at least 20 different measures to quantify (un)fairness have been proposed [5]. Different fairness measures often exhibit complex relationships among them [2], [3], [4], [5], [6], [7], [8], such as conflicts, inconsistencies, or even unknown patterns. Additionally, fairness is often conflicting with the accuracy of learning models [2], [3], [4]. Thus, the intricate relationships among accuracy and multiple fairness measures pose great challenges in fair machine learning and fair artificial intelligence in general.

Various techniques have been developed to optimize fairness measures of learning models, which can be mainly divided into two categories [2], [3], [4]. The core idea of the first category converts accuracy and multiple fairness measures into one combined objective to be optimized. One such technique is multi-FR [9], which calculates the losses of all the measures and uses a weighted sum of those losses to update a learning model. Multiobjective evolutionary learning (MOEL) [10], as the second category, demonstrates significant advantages in training fairer machine learning models [11], [12], [13], [14], [15]. In this category, a learning algorithm operates by explicitly defining a set of measures, including accuracy and multiple fairness measures, and simultaneously optimizing these measures during the training process, where each measure is viewed as an objective [2], [3], [4]. MOEL can generate a diverse set of learning models, with each model representing a tradeoff among different measures.

Optimizing all the fairness measures may not always be necessary. Recent studies [4], [7], [16] show that a comprehensive set of fairness measures can be represented by a subset because of the positive correlations among some measures. Such a subset is denoted as a *representative measure subset*. In the light of this, the work of [12] proposed to optimize a representative subset of measures [16] throughout the model training. The findings of [12] show that by optimizing this carefully selected subset, improvements can be achieved across all the fairness measures, even including those that were not used as the optimization objectives during the model training.

However, using a static predefined measure subset [12] still has limitations for three reasons. First, the prior knowledge or significant computational costs were involved in finding a subset that can comprehensively represent all the measures [7], [16]. Second, the proper representative subset may vary across different datasets. Carefully determining the representative subset of measures for specific datasets requires extra computational cost before model training. Third, the correlation among accuracy and multiple fairness measures is changing along with the model training stages. The optimal representative subset of measures at one optimization stage may not necessarily remain optimal at another stage.

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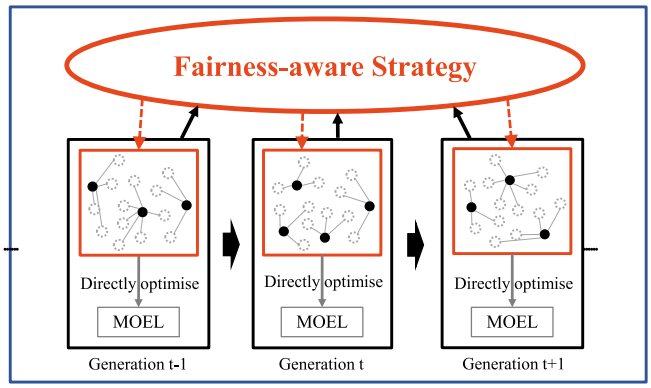


Fig. 1. Flow of our framework, where a fairness-aware strategy is used to dynamically select a representative subset (the solid circles) to be optimized using MOEL at each generation to improve all the measures (all the circles). Each generation involves mating selection, reproduction, and survival selection encompassing the loop outlined in lines 6–13 of Algorithm 1, which will be detailed in Section III-A.

Because of the aforementioned issues, instead of using a static predefined one, an adaptively online-determined representative subset that does not need any prior knowledge is more promising as the optimizing objectives during model training. Novel contributions of this article are as follows.

1) We introduce a fairness-aware strategy assisted MOEL (FaMOEL) framework to optimize an objective set, including accuracy and multiple fairness measures as shown in Fig. 1. Along with the training process of FaMOEL, our framework is aware of the current model training process and then adaptively determines a representative subset (solid circles in Fig. 1) of measures as objectives instead of using a predefined static one. Thus, the representative subset may vary along with the model training process.
2) Based on our framework, an efficient instantiation algorithm is developed. Specifically, we design and incorporate three enhancement strategies into ORNCIE [17], aiming to construct the most appropriate representative measure subset.

We demonstrate the effectiveness of our framework and its instantiation on the 12 benchmark datasets. Empirical results observed by four performance indicators reveal that our framework achieves outstanding performance in terms of accuracy and 25 fairness measures compared to the state-of-the-art methods.[1] Our framework can well perceive a suitable representative subset according to the current model training process. The results also demonstrate that the most appropriate representative subset did vary from generation to generation.

The remainder of this article is organized as follows. Section II introduces the background. Our proposed framework FaMOEL and the designed algorithm based on this framework are presented in Section III. Section IV presents and discusses the experimental studies. Section V concludes this article and discusses the future work.

[1]Code of this work is available at https://github.com/qingquan63/FaMOEL.

TABLE I
CONFUSION MATRIX

	Actual positive	Actual negative
Predicted positive	TP PPV = TP/(TP+FP) TPR = TP/(TP+FN)	FP FDR = FP/(TP+FP) FPR = FP/(FP+TN)
Predicted negative	FN FOR = FN/(TN+FN) FNR = FN/(TP+FN)	TN NPV = TN/(TN+FN) TNR = TN/(TN+FP)

II. BACKGROUND

This section presents an overview of fairness measures and their relationship. Then, the MOEL framework for fairer machine learning is introduced.

A. Fairness Measures in Machine Learning

Numerous measures have been proposed to evaluate (un)fairness from ethical standpoints in the context of fairness [2], [3], [4], [5], [6], [7], [8]. There is no consensus on an universally agreed fairness measure that is capable of comprehensively taking into account all the perspectives of fairness. Many measures exhibit diverse and intricate relationships with one another. Some measures demonstrate positive correlations with others, while some others present conflicts. Additionally, accuracy and some fairness measures are often conflicting or inconsistent with each other [5], [7].

Based on the confusion matrix (shown in Table I) or other principles, the study of [7] comprehensively reviews 25 fairness measures, which are detailed in Table II. Given a sensitive attribute (i.e., gender and race), a dataset can be divided into two groups, unprivileged group g_u and privileged group g_p. The fairness measures Fairs 1–13 are formulated based on the confusion matrix of groups g_u and g_p. In Fairs 6 and 11, *ERR* is equal to $(\mathrm{FN}+\mathrm{FP})/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN}+\mathrm{TN})$. G, y, and $\hat{y}$ denote the sensitive attributes, true labels, and predicted labels obtained by a learning model, respectively. In Fairs 16–24, $|G|$ is the number of groups, n_g refers to the size of group g, n is the number of observations (i.e., $n = \sum_g n_g$), and α is a positive constant. The benefit vector b_i is equal to $\hat{y}_i - y_i + 1$ for the ith data. $b_i^{g_u}$ and $b_i^{g_d}$ are the benefit values of the ith data in unprivileged and privileged groups, respectively. μ, μ_{g_u}, and μ_{g_d} are the mean values of all the b_i, $b_i^{g_u}$, and $b_i^{g_d}$, respectively.

The comprehensive analysis in the work of [7] provides insights into the relationships and tradeoffs among these 25 fairness measures. It suggests that it is possible to cluster all those fairness measures into six groups based on their correlation [7]. Specifically, the conflicts and consistencies among these measures are analysed through a number of trained models obtained by three algorithms, including the logistic regression, a reweighing method [18], and a meta fair method [19], across seven datasets. Then, the obtained six groups can help to construct a representative measure subset. Note that, this process needs significant computational cost [7]. However, some fairness measures are positively correlated in one dataset, while they are negatively correlated in the other datasets [6], [16]. Therefore, a proper representative subset for one dataset may not be suitable for the other datasets.

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TABLE II
SUMMARY OF 25 FAIRNESS MEASURES [7]

Notation	Name	Formulation
Fair1	True positive rate difference	$TPR(g_u) - TPR(g_p)$
Fair2	False positive rate difference	$FPR(g_u) - FPR(g_p)$
Fair3	False negative rate difference	$FNR(g_u) - FNR(g_p)$
Fair4	False omission rate difference	$FOR(g_u) - FOR(g_p)$
Fair5	False discovery rate difference	$FDR(g_u) - FDR(g_p)$
Fair6	Error rate difference	$ERR(g_u) - ERR(g_p)$
Fair7	False positive rate ratio	$FPR(g_u)/FPR(g_p)$
Fair8	False negative rate ratio	$FNR(g_u)/FNR(g_p)$
Fair9	False omission rate ratio	$FOR(g_u)/FOR(g_p)$
Fair10	False discovery rate ratio	$FDR(g_u)/FDR(g_p)$
Fair11	Error rate ratio	$ERR(g_u)/ERR(g_p)$
Fair12	Average odds difference	$\frac{1}{2}(TPR(g_u) - TPR(g_p) + FPR(g_u) - FPR(g_p))$
Fair13	Average abs odds difference	$\frac{1}{2}\left(\lvert TPR(g_u) - TPR(g_p)\rvert + \lvert FPR(g_u) - FPR(g_p)\rvert\right)$
Fair14	Disparate impact	$P(\hat{y}=1\lvert G=g_u)/P(\hat{y}=1\lvert G=g_p)$
Fair15	Statistical parity difference	$P(\hat{y}=1\lvert G=g_u) - P(\hat{y}=1\lvert G=g_p)$
Fair16	Generalized entropy index	$\frac{1}{n\alpha(\alpha-1)}\sum_{i=1}^{n}\left[\left(\frac{b_i}{\mu}\right)^{\alpha}-1\right]$
Fair17	Between all groups generalized entropy index	$\frac{1}{n\alpha(\alpha-1)}\sum_{g=1}^{\lvert G\rvert} n_g\left[\left(\frac{\mu_g}{\mu}\right)^{\alpha}-1\right]$
Fair18	Between group generalized entropy index	$\frac{1}{n\alpha(\alpha-1)} n_{g_u}\left[\left(\frac{\mu_{g_u}}{\mu}\right)^{\alpha}-1\right] + \frac{1}{n\alpha(\alpha-1)} n_{g_d}\left[\left(\frac{\mu_{g_d}}{\mu}\right)^{\alpha}-1\right]$
Fair19	Theil index	$\frac{1}{n}\sum_{i=1}^{n}\frac{b_i}{\mu} ln\frac{b_i}{\mu}$
Fair20	Coefficient of variation	$2\sqrt{\frac{1}{n}\sum_{i=1}^{n}\frac{b_i}{\mu} ln\frac{b_i}{\mu}}$
Fair21	Between group theil index	$\frac{1}{n_{g_u}}\sum_{i=1}^{n_{g_u}}\frac{b_i^{g_u}}{\mu_{g_u}} ln\frac{b_i^{g_u}}{\mu_{g_u}} + \frac{1}{n_{g_p}}\sum_{i=1}^{n_{g_p}}\frac{b_i^{g_p}}{\mu_{g_p}} ln\frac{b_i^{g_p}}{\mu_{g_p}}$
Fair22	Between group coefficient of variation	$2\sqrt{\frac{1}{n_{g_u}}\sum_{i=1}^{n_{g_u}}\frac{b_i^{g_u}}{\mu_{g_u}} ln\frac{b_i^{g_u}}{\mu_{g_u}}} + 2\sqrt{\frac{1}{n_{g_p}}\sum_{i=1}^{n_{g_p}}\frac{b_i^{g_p}}{\mu_{g_p}} ln\frac{b_i^{g_p}}{\mu_{g_p}}}$
Fair23	Between all groups theil index	$\sum_{g=1}^{\lvert G\rvert}\frac{1}{n_g}\sum_{i=1}^{n_g}\frac{b_i^g}{\mu_g} ln\frac{b_i^g}{\mu_g}$
Fair24	Between all groups coefficient of variation	$\sum_{g=1}^{\lvert G\rvert} 2\sqrt{\frac{1}{n_g}\sum_{i=1}^{n_g}\frac{b_i^g}{\mu_g} ln\frac{b_i^g}{\mu_g}}$
Fair25	Differential fairness bias amplification	Difference in smoothed empirical differential fairness between the classifier and the original dataset [20]

B. Mitigating Unfairness Through Multiobjective Evolutionary Learning

MOEL [10] has been proposed to optimize accuracy and multiple fairness measures for fairer machine learning [8], [11], [12], [13], aiming to evolve a population of learning models, e.g., artificial neural nets (ANNs), by utilizing multiobjective evolutionary algorithms (MOEAs) [21]. MOEAs represent a class of optimization techniques specifically designed to address problems with multiple conflicting objectives. Unlike the traditional single-objective optimization methods, a set of optimal solutions, called optimal Pareto front is desired when solving multiobjective optimization problems [22]. MOEAs typically aim to approximate the optimal Pareto front by maintaining a set of solutions. When evaluating the solution set, four aspects are considered, namely convergence, spread, uniformity, and cardinality [22].

In mitigating unfairness, MOEL [8], [11], [12], [13] can provide a diverse model set, where a model in the set indicates a tradeoff among the accuracy and different fairness measures. Later, study [12] constructs an ensemble model with the diverse model set to automatically balance accuracy and fairness measures. In the MOEL-based algorithms [11], [12], it is worth noting that the objectives considered, which include accuracy and fairness measures, may not always be differentiable. Furthermore, study [13] confirms that leveraging an appropriate gradient from adversarial learning can significantly enhance model fairness when performing partial training [23], [24]. However, this method is limited to optimizing only two fairness measures: 1) equalized odds and 2) demographic parity.

Given a set of fairness measures, it is not always necessary to optimize all of them thanks to positive correlations among some measures [4], [7], [16]. Particularly, the studies of [16] analyse the relationship among many fairness measures and select representative measures. Motivated by this observation, the study of [12] uses accuracy and the selected representative fairness measures [16] as the optimization objectives in MOEL. Note that throughout the entire model training process, even on different datasets, the study [12] only optimizes those predefined measures.

However, as mentioned before, each dataset may have a different suitable representative fairness subset [6], [16]. Determining a suitable representative fairness subset for a new dataset consumes significant computational cost. Furthermore, even when dealing with the same dataset in one model training trial, the suitable representative subset may vary across different training stages.

III. FAIRNESS-AWARE MULTIOBJECTIVE EVOLUTIONARY LEARNING

Section III-A presents our proposed framework, namely FaMOEL, which dynamically and adaptively determines a representative subset of fairness measures during model training without any prior knowledge. The determined measures

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Algorithm 1 Fairness-Aware Multiobjective Learning Framework

Input: Initial models $\mathcal{M}_1, \ldots, \mathcal{M}_\lambda$, set of model evaluation objectives $\mathcal{E}$, training dataset $\mathcal{D}_{train}$, validation dataset $\mathcal{D}_{validation}$, multiobjective optimiser π, fairness-aware strategy *FA*

Output: A final model set $\mathcal{M}_1, \ldots, \mathcal{M}_\lambda$

1: Partially train [23], [24] $\mathcal{M}_1, \ldots, \mathcal{M}_\lambda$ over $\mathcal{D}_{train}$
2: **for** $i \in \{1, \ldots, \lambda\}$ **do**
3: $\epsilon_i \leftarrow$ Evaluate $\mathcal{M}_i$ with objectives $\mathcal{E}$ on $\mathcal{D}_{validation}$
4: **end for**
5: **while** terminal conditions are not fulfilled **do**
6: $\mathcal{E}' \leftarrow$ Perform fairness-aware *FA* to select
7: representative objectives from $\mathcal{E}$
7: $\mathcal{P} \leftarrow$ Select μ promising models from $\mathcal{M}_1, \ldots, \mathcal{M}_\lambda$
8: with "best" $\epsilon_1, \ldots, \epsilon_\mu$ according to π and $\mathcal{E}'$
8: $\mathcal{M}' \leftarrow$ Generate ϕ new models $\mathcal{M}'_1, \ldots, \mathcal{M}'_\phi$ from $\mathcal{P}$
9: according to π
9: **for** $i \in \{1, \ldots, \phi\}$ **do**
10: $\mathcal{M}'_i \leftarrow$ Partially train [23], [24] $\mathcal{M}'_i$ on $\mathcal{D}_{train}$
11: $\epsilon'_i \leftarrow$ Evaluate $\mathcal{M}'_i$ with objectives $\mathcal{E}$ on $\mathcal{D}_{validation}$
12: **end for**
13: $< \mathcal{M}_1, \epsilon_1 >, \ldots, < \mathcal{M}_\lambda, \epsilon_\lambda > \leftarrow$ Select λ promising
14: models from $\{\mathcal{M}_1, \ldots, \mathcal{M}_\lambda\} \bigcup \{\mathcal{M}'_1, \ldots, \mathcal{M}'_\phi\}$ by π
14: and $\mathcal{E}'$ based on $\epsilon_1, \ldots, \epsilon_\lambda$ and $\epsilon'_1, \ldots, \epsilon'_\phi$, and then
14: update $\mathcal{M}_1, \ldots, \mathcal{M}_\lambda$ and $\epsilon_1, \ldots, \epsilon_\lambda$ accordingly
14: **end while**

are used as objectives of MOEL to guide the evolution of learning models. Then, an instantiation algorithm based on FaMOEL is implemented in Section III-B, where three enhancement strategies are designed to improve the fairness-awareness ability of our method.

A. Fairness-Aware Multiobjective Evolutionary Learning Framework for Mitigating Unfairness

First, in our study, we formulate the task of improving the learning model's accuracy and fairness as a multiobjective learning task [8], [11], [12], [13]

$$\text{minimise}_{\mathbf{x}\in\Omega} \quad F(\mathbf{x}) = \{f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_M(\mathbf{x})\} \tag{1}$$

where $\mathbf{x}$ represents the parameters of a learning model within the decision space Ω. $F(\mathbf{x})$ is a set of M objective functions that assess the accuracy and fairness of the model parametrized by $\mathbf{x}$ on the given task.

Algorithm 1 outlines our proposed FaMOEL with six inputs, including an initial population of models $\mathcal{M}$, a set of model evaluation objectives $\mathcal{E}$, a fairness-aware strategy *FA*, the training data $\mathcal{D}_{\text{train}}$, the validation data $\mathcal{D}_{\text{validation}}$, and a multiobjective optimizer π. The objectives in $\mathcal{E}$ are used to calculate optimized objective values, such as accuracy and fairness measures based on the predictions of the models in $\mathcal{M}$ on the validation data $\mathcal{D}_{\text{validation}}$. Fairness-aware strategy *FA* is used to find a representative subset from the entire objectives $\mathcal{E}$ during the model training. Note that, compared with the previous work [11], as clearly illustrated in Fig. 1, the core difference is the fairness-aware strategy *FA*. The training data $\mathcal{D}_{\text{train}}$ is utilized for the local search strategies, such as partial training [23], [24], to update the parameters of the models in $\mathcal{M}$.

The multiobjective optimizer π consists of three main strategies [11], [12]: 1) reproduction; 2) mating selection; and 3) survival selection. In our framework, during model initialization and generation, partial training [23], [24] is always applied to the models using the training data $\mathcal{D}_{train}$. The objective values of each model are obtained through the evaluation objectives $\mathcal{E}$ (lines 1 and 6 in Algorithm 1). In the main loop, the fairness-aware strategy *FA* is performed to online select a representative subset $\mathcal{E}'$ from the evaluation objectives $\mathcal{E}$ according to the current evolution process (line 6 in Algorithm 1). Then, the mating selection strategy of π selects a promising set of parent models $\mathcal{P}$ from the population $\mathcal{M}$ (line 7 in Algorithm 1) only considering $\mathcal{E}'$. After that, ϕ new models, denoted as $\mathcal{M}'$, are generated by inheriting information from $\mathcal{P}$ through the reproduction strategy of π (line 8 in Algorithm 1). This strategy modifies the parameters of the parent models, often using operators like crossover and mutation. After partial training and model evaluation, λ candidate models are selected from the combination of the original population $\mathcal{M}$ and the new models $\mathcal{M}'$ using the survival selection strategy of π considering the representative objectives $\mathcal{E}'$ (line 13 in Algorithm 1). These selected models form the updated population $\mathcal{M}$ for the next generation. These steps are repeated until a termination criterion is met. Finally, a model set $\mathcal{M}_1, \ldots, \mathcal{M}_\lambda$ is obtained, from which decision makers can select one or multiple to deploy according to specific requirements in real-world scenarios.

B. Instantiation Algorithm Based on Our Framework

To verify the effectiveness of our framework FaMOEL, an instantiation algorithm based on FaMOEL is developed and the key components of FaMOEL, including the model set, evaluation objectives, the fairness-aware method, and the multiobjective optimization algorithm are introduced as follows. Noted that, our proposed framework allows for flexibility in selecting these components based on specific prediction tasks and preferences.

1) Model Set: A range of machine learning models can be utilized within our framework. In our study, a collection of ANNs with the same architecture is employed as individuals. Each ANN's weights and biases are encoded as a real-value vector and represented as an individual [12], [23].

2) Evaluation Objectives: In this study, a total of 26 fairness measures, including accuracy and Fair 1 to Fair 25 (as listed in Table II), are considered. The accuracy is evaluated using the cross-entropy (CE) measure commonly employed for the classifiers [12], and it is minimized. Following [12], the absolute values of Fairs 1–6, Fair 12, Fair 13, and Fair 15 are minimized. For Fairs 7–11 and Fair 14 using ratios, we construct the objective functions to be minimized with the transformation following the work of [12]. Taking Fair 7 ($\text{FPR}(g_u)/\text{FPR}(g_p)$) as an

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example, its corresponding objective function is calculated as $1 - \min\{[\text{FPR}(g_u)/\text{FPR}(g_p)], [\text{FPR}(g_p)/\text{FPR}(g_u)]\}$. Fairs 16–25 are directly used as objective values to be minimized and their values are always positive. The transformed objectives corresponding to Fairs 1–25 are denoted as f_1–f_{25}, respectively. The optimal values of f_1–f_{25} are all zeros.

3) Multiobjective Optimiser: Parent selection, survival selection and reproduction strategy of π can be implemented by any MOEA. In our instantiation algorithm, we utilize Two_Arch2 [25] for both the parent selection and survival selection. The survey [26] demonstrates the efficacy of Two_Arch2 to address many-objective optimization. Two_Arch2 is popular and efficient [21], [26], [27], [28], exhibiting competitive performance in handling many-objective optimization problems, which maintains two archives, each focusing on convergence and diversity of individuals, respectively. In the reproduction strategy, isotropic Gaussian perturbation and the variant of weight crossover are applied as mutation and crossover operators [12], respectively. A set of new $\mathcal{M}'$ (line 10 in Algorithm 1) can be obtained by applying the mutation operator to the model set resulting from the crossover between the convergence archive and the diversity archive [25]. Specifically, the isotropic Gaussian perturbation [12] is performed as the mutation operator, formulated as $r_i = r_i + \delta_i$, where the ith weight of an ANN, denoted as r_i, undergoes the isotropic Gaussian perturbation, with δ_i sampled from a normal distribution $\mathcal{N}(0, \sigma^2)$. σ represents the mutation strength. Given parents p and q, the weight crossover [12] is applied as $r_i^{o_1} = u_i r_i^p + (1-u_i) r_i^q$ and $r_i^{o_2} = u_i r_i^q + (1-u_i) r_i^p$, where u_i is sampled from (0, 1) uniformly at random. Meanwhile, r_i^p, r_i^q, $r_i^{o_1}$, and $r_i^{o_2}$ denote the ith weight of the parent p, parent q, offspring o_1, and offspring o_2, respectively.

Regarding the partial training [12], [23], [24], the model parameters are updated by the stochastic gradient descent (SGD) optimizer [29].

4) Fairness-Aware Strategy: The fairness-aware strategy is to adaptively construct a representative subset from all the objectives to be considered according to the current evolution status of each generation. In the literature, objective subset selection methods can be generally divided into three categories [30]: 1) dominance-based; 2) model-based; and 3) correlation-based methods. Dominance-based methods [31], [32] aim to identify representative objectives that preserve the dominance structure as much as possible. In contrast, the model-based methods [30], [33] build a model to approximate the obtained nondominated front and select representative objectives based on the model's coefficients. However, both the dominance-based and model-based methods are less effective for the problems with many objectives (up to 15) [30]. Dominance-based methods struggle to maintain the informative dominance structure when there is a high proportion of nondominated solutions [32]. The models constructed by the model-based methods often lack accuracy due to the relatively limited number of nondominated solutions in a highly approximated space with many objectives [30]. In contrast, the correlation-based methods [17], [34] select representative objectives by leveraging the correlation relationships among the objectives. Among these, ORNCIE [17] has proven to be effective even in problems with up to 50 objectives. Therefore, we have chosen ORNCIE as the fairness-aware strategy in our study.

Algorithm 2 Fairness-Aware Strategy

Input: Current generation t, set of model evaluation objectives $\mathcal{E}$, history of mNCIE matrices $NC = \{NC_1, \ldots, NC_{t-1}\}$, objective values of current population ϵ, selection threshold τ

Output: Set of representative objectives $\mathcal{E}'$, history of mNCIE matrices NC

1: $NC_t \leftarrow$ Calculate the mNCIE matrix [17] according to the objective values of current population ϵ
2: $NC = NC \bigcup \{NC_t\}$
3: **if** $t < 10$ **then**
4: $\quad \mathcal{E}' = \mathcal{E}$
5: **else**
6: $\quad NC^r \leftarrow$ Calculate an average mNCIE matrix
7: $\quad$ according to the matrices NC of the last 10 generations
7: $\quad S = [1, 2, \ldots, |\mathcal{E}|]$
8: $\quad \mathcal{E}' = \emptyset$
9: $\quad$ **while** $S \neq \emptyset$ **do**
10: $\quad\quad$ **if** all the elements in NC^r are positive **then**
11: $\quad\quad\quad J = argmax_j(sum(NC^r(1:|\mathcal{E}|, j)))$, where
12: $\quad\quad\quad j \in S$
12: $\quad\quad$ **else**
13: $\quad\quad\quad J = argmin_j(sum(NC^r(i, j)))$, where
14: $\quad\quad\quad NC^r(i, j) < 0,\ 1 \leq i \leq m$ and $j \in S$
14: $\quad\quad$ **end if**
15: $\quad\quad S = S/\{J\}$
16: $\quad\quad \mathcal{E}' = \mathcal{E}' \bigcup \{\mathcal{E}_J\}$
17: $\quad\quad Del = \{j | NC^r(J, j) > \tau\}$, where $j \in S$
18: $\quad\quad S = S/Del$
19: $\quad$ **end while**
20: **end if**

Inspired by ORNCIE [17], we propose our fairness-aware strategy shown in Algorithm 2, by adding three novel enhancement strategies based on ORNCIE. ORNCIE calculates a modified nonlinear correlation information entropy (mNCIE) [17] to analyse the interrelationships among the objectives according to the current population information and subsequently identify a representative subset. The obtained subset is directly optimized by a multiobjective optimizer, such as Two_Arch2 [25].

As described in Algorithm 2, our proposed fairness-aware strategy takes five inputs, including current generation t, set of model evaluation objectives $\mathcal{E}$, history of mNCIE matrices $NC = \{NC_1, \ldots, NC_{t-1}\}$, selection threshold τ, and objective values of current population ϵ (obtained from Algorithm 1). First, mNCIE NC_t [17] is calculated according to ϵ (line 1 in Algorithm 2). Each matrix in NC is a symmetry matrix with a size of $|\mathcal{E}| \times |\mathcal{E}|$, where each value in the matrix falls within the range of $[-1, 1]$. A large positive value between a pair of objectives indicates that the two objectives are highly

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positively correlated. On the contrary, a low negative value between a pair of objectives suggests a strong negative correlation between them. Then, NC_t is appended to the historical mNCIE matrices (line 2 in Algorithm 2). If the generation number t is less than 10, the entire objective set $\mathcal{E}$ is used as objectives to be optimized (line 4 in Algorithm 2), which is viewed as warm starting. The warm starting aims to enhance the exploration ability of the evolution in the early stage by considering the entire $\mathcal{E}$, which is the first enhancement. After ten generations, NC_r is calculated by taking into account the final ten matrices in NC (line 1 in Algorithm 2), that is $NC_{t-9}, \ldots, NC_t$. Here, $NC^r(i,j) = 0.1\sum_{k=0}^{9} NC_{t-9+k}(i,j)$, where $i, j \in [1, |\mathcal{E}|]$. In the original ORNCIE, only NC_t is considered to detect the training process rather than the NC^r we used. This second enhancement is because NC^r can more precisely capture the correlations among the objectives for the current model training.

In the loop, the algorithm identifies the most essential (here defined as conflicting) objective, denoted as $\mathcal{E}_J$. Subsequently, the objectives, referred to as *Del*, that are positively correlated with $\mathcal{E}_J$ are excluded (line 17 in Algorithm 2). When determining *Del*, the original approach [17] uses the "classifying objectives" strategy (shown in Algorithm 2 of this article [17]). Instead, we use a static hyperparameter τ to identify *Del*, which is the third enhancement strategy. The dynamic determination *Del* using the classifying objectives strategy may wrongly remove some essential objectives and further degrade the representation ability of $\mathcal{E}'$. Following this iterative process, a representative objective subset $\mathcal{E}'$ is determined. The appropriate setting of the unique hyperparameter τ will be discussed in Section IV-E.

IV. Experimental Studies

In this section, Section IV-A introduces the aims of our experimental studies. Then, Section IV-B presents the experimental settings. Four experiments are presented and discussed in Section IV-C to Section IV-G, respectively.

A. Overview of Experimental Studies

Four experiments are conducted to achieve a comprehensive analysis of our proposed fairness-aware MOEL framework and its instantiation algorithm. Experimental setting is detailed in Section IV-B. First, we verify the effectiveness of our fairness-aware framework in Section IV-C by comparing it with the two state-of-the-art methods in multiobjective optimization for fair machine learning: one that optimizes the entire set of objectives [11] and another that uses a static representative subset [12]. Next, in Section IV-D, to further investigate the capabilities of our framework, we analyse whether the frequently selected objectives obtained by our algorithm for each dataset are more suitable as a new representative subset. Then, in Section IV-E, ablation studies to assess the effectiveness of our fairness-aware strategy are conducted. Furthermore, Section IV-F discusses the computational cost of our method using the fairness-aware framework. Finally, in Section IV-G analyses the sensitivity of the unique parameter τ of our method as described in Algorithms 1 and 2, and suggests a value for τ.

B. Experimental Setting

1) Compared Methods: A total of 26 objectives, including accuracy *CE* and 25 fairness measures (f_1–f_{25}, described in Table II), are considered for all the methods. Our proposed method, denoted as *FaMOEL*, is described in Algorithms 1 and 2. We compare *FaMOEL* with the two state-of-the-art algorithms, namely *MOEL* [11] and MOEL$_{\text{Rep}}$ [12], respectively. *MOEL* directly optimizes 26 objectives through the MOEL framework. MOEL$_{\text{Rep}}$ focuses on optimizing a static representative subset. The work of [7] clusters f_1–f_{25} into six groups. The static representative subset can be identified by selecting a measure from each group. Our study selects f_4, f_7, f_{10}, f_{16}, f_{17}, and f_{25} as the representative fairness measure subset. Thus, MOEL$_{\text{Rep}}$ optimizes *CE* and these six fairness objectives.

2) Datasets: Twelve well-known benchmark datasets widely used in the literature of fairness [7], [35] are considered, namely Heart health [36], Titanic [37], German [38], Student performance [39], COMPAS [40], Bank [41], Adult [42], Drug [43], Patient [44], LSAT [45], Default [46], and Dutch [18]. Table III summarises these datasets used. Note that, the analysis of the static representative objective subset for MOEL$_{\text{Rep}}$ is based on the first seven datasets [7], while the remaining five datasets are not included. This can help to further verify the effectiveness of our framework on the new benchmark datasets. The preprocessing steps for Heart health, Titanic, German, Student performance, COMPAS, Bank, and Adult datasets follow the same procedure as described in [7]. Each dataset is randomly divided into three partitions: 1) a training set; 2) a validation set; and 3) a test set with a split ratio of 6:2:2. All the sensitive features listed in Table III are taken into consideration in the calculation of these objectives.

3) Parameter Settings: All the methods, including *FaMOEL*, MOEL$_{\text{Rep}}$, and *MOEL* use the same settings for a fair comparison, introduced as follows. Each individual in the population is designed with a fully connected architecture, consisting of one hidden layer [11], [12]. The values of the learning rate, mutation strength, and the number of hidden nodes are determined using the grid search and are summarized in Table IV. The multiobjective optimizer π is Two_Arch2 [25]. The size of the convergence and diversity archives are all set as 100. We set the termination condition to a maximum of 100 generations. In our fairness-aware method *FaMOEL*, the selection threshold τ is set to 0.22. Five-fold cross-validation is used in our experiments, where ten trials are independently run for each fold. Thus, 50 trials in total are performed for each benchmark dataset.

4) Performance Measures: The quality of a model set can be assessed from the four aspects, including convergence, spread, uniformity, and cardinality [22]. For a more comprehensive analysis, we adopt the widely used generational distance (GD) [47] for convergence, pure diversity (PD) [48] for spread, and spacing (SP) [49] for uniformity as suggested in [22] and [50], and summarize them in Table V. PD and

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TABLE III
12 BENCHMARK DATASETS USED IN OUR STUDY

Dataset	Source	Domain	Sensitive	{Privileged, Unprivileged}	Description of Prediction Task	Imbalance Rate(+:-)
Heart health	Faisalabad Institute of Cardiology and Allied Hospital in Faisalabad	Healthcare	Age	{Young, Old}	Whether a person will have heart disease or not	1.17:1
Titanic	Titanic disaster passenger list	Disaster	Gender	{Male, Female}	Whether a person will survive sinking of Titanic or not	1:1.61
German	-	Finance	Gender	{Male, Female}	Whether a person has an acceptable credit risk or not	2.33:1
Student performance	Two Portuguese secondary schools	Education	Gender	{Male, Female}	Whether a student will pass the exam or not	1:1.57
COMPAS	The Broward County record	Criminology	Gender	{Male, Female}	Whether an arrested offender will be rearrested within two years counting from taking the test or not	1:1.20
Bank	A Portuguese banking institution direct marketing campaigns	Finance	Age	{Old, Young}	Whether a client will subscribe to a term or not	1:7.55
Adult	Census bureau database	Finance	Gender	{Male, Female}	Whether a person can get income higher or not	1:3.03
Drug consumption	Rampton Hospital online survey	Healthcare	Gender	{Female, Male}	Whether a person never used mushroom before or not	1.72:1
Patient treatment	An Indonesia private hospital	Healthcare	Gender	{Male, Female}	Whether a patient will be in care or not	1.52:1
LSAT	An American Law School Admission Council survey across 163 law schools	Education	Gender	{Male, Female}	Whether a student will pass the exam or not	8.07:1
Default	Chinese (Taiwan) customers' default payments	Finance	Gender	{Male, Female}	Whether a customer will default on payments or not	1:7.55
Dutch	Dutch aggregated groups	Finance	Gender	{Male, Female}	Whether a person has a highly prestigious or not	1:1.10

TABLE IV
PARAMETER SETTINGS OF ALGORITHMS

Dataset	Learning Rate	Mutation Strength	#Nodes
Heart health	0.0001	0.0001	16
Titanic	0.001	0.0001	8
German	0.0001	0.05	64
Student performance	0.001	0.0001	64
COMPAS	0.001	0.05	64
Bank	0.001	0.005	64
Adult	0.001	0.05	64
Drug consumption	0.001	0.0001	64
Patient treatment	0.0001	0.0001	64
LSAT	0.001	0.005	64
Default	0.001	0.01	64
Dutch	0.001	0.01	64

TABLE V
QUALITY INDICATORS FOR EVALUATING A SOLUTION SET

Quality indicator	Convergence	Spread	Uniformity	Cardinality
Generational distance (GD)	✓			
Pure diversity (PD)		✓		
Spacing (SP)			✓	
Hypervolume (HV)	✓	✓	✓	✓

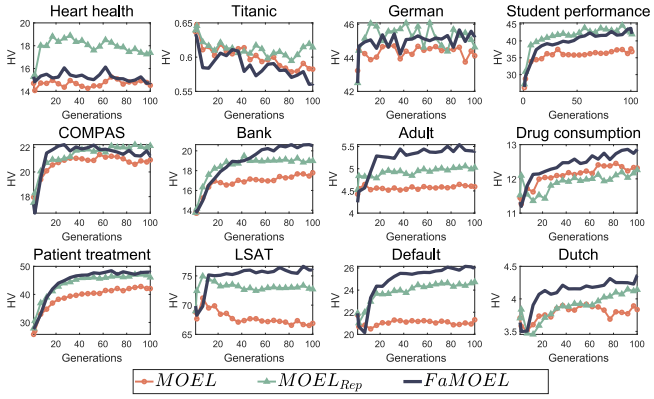


Fig. 2. HV curves along with generations averaged over 50 trials considering accuracy and f_1–f_{25}.

SP can collectively depict the diversity of a solution set. A solution set with diverse performance not only provides decision makers with a better understanding of the task at hand but also offers flexibility, allowing them to choose the most suitable ones according to varying requirements. Moreover, hypervolume (HV) [51], also known as the only indicator with the Pareto compliance [52], is applied to measure the overall performance. All the objective values are normalized before computing HV values. The reference point $(1.2, 1.2, \ldots, 1.2)$ is used for HV. A larger PD or HV value indicates better performance with respect to its corresponding property, while a smaller GD or SP value indicates better performance.

C. Effectiveness of Our Framework

To verify the effectiveness of our fairness-aware MOEL framework, three perspectives are considered: 1) convergence curves of HV values of *MOEL*, MOEL_{Rep}, and our proposed *FaMOEL*; 2) quality of population of the final generation in terms of GD, PD, SP, and HV; and 3) visualization of the fairness-aware process of our proposed *FaMOEL*, Fig. 2 illustrates the convergence curves of the HV values considering *MOEL*, MOEL_{Rep}, and our proposed *FaMOEL* on the test data. In the calculation of HV, the pseudo Pareto front is the nondominated model set with respect to 26 objectives by considering all the models obtained by the three compared algorithms from all the generations across 50 trials as suggested in [12]. Fig. 2 reveals that our proposed method *FaMOEL* (the black) is better than *MOEL* (the orange) and MOEL_{Rep} (the green) on 7 out of 12 datasets, including Bank, Adult, Drug consumption, Patient treatment, LSAT, Default, and Dutch. It means that *FaMOEL* can achieve better overall performance in terms of *CE* and f_1–f_{25} and outperforms the state-of-the-art algorithms on these seven datasets, resulting from the fairness-aware strategy. The poor performance of *FaMOEL* on the remaining five datasets may be partially attributed to the small size [53] of the datasets. A closer examination is expected in the future.

The curves of MOEL_{Rep} exhibit an increasing trend with the generations on 7 out of 12 datasets, including

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TABLE VI
GD, PD, SP, AND HV VALUES OF FINAL MODEL SET AVERAGED OVER 50 TRIALS. "+/≈/−" INDICATES THAT THE AVERAGE INDICATOR VALUE OF THE CORRESPONDING ALGORITHM (SPECIFIED BY COLUMN HEADER) IS STATISTICALLY BETTER/SIMILAR/WORSE THAN THE ONE OF *FaMOEL* ACCORDING TO THE FRIEDMAN TEST WITH A 0.05 SIGNIFICANCE LEVEL. THE BEST AVERAGED VALUES ARE HIGHLIGHTED IN GRAY

Dataset	GD			PD		
	MOEL	$MOEL_{Rep}$	*FaMOEL*	*MOEL*	$MOEL_{Rep}$	*FaMOEL*
Heart health	2.96e-04(8.9e-05)≈	2.56e-04(8.6e-05)+	2.88e-04(8.3e-05)	2.31e+14(4.7e+13)+	1.45e+14(5.0e+13)-	2.08e+14(4.5e+13)
Titanic	1.36e-04(8.7e-05)-	1.28e-04(8.3e-05)≈	1.25e-04(8.8e-05)	4.72e+14(3.8e+13)+	4.31e+14(3.3e+13)≈	4.40e+14(4.0e+13)
German	5.66e-04(2.2e-04)-	4.85e-04(1.6e-04)≈	5.24e-04(2.1e-04)	4.11e+14(1.0e+14)+	3.71e+14(1.2e+14)-	3.92e+14(1.1e+14)
Student performance	6.11e-04(1.7e-04)-	4.66e-04(1.2e-04)+	5.04e-04(1.5e-04)	4.51e+14(7.7e+13)+	3.77e+14(7.7e+13)-	4.06e+14(9.3e+13)
COMPAS	2.09e-04(6.7e-05)-	1.87e-04(5.7e-05)+	2.00e-04(6.6e-05)	4.24e+14(5.0e+13)+	3.99e+14(4.7e+13)≈	4.07e+14(4.6e+13)
Bank	9.41e-05(8.3e-05)-	7.94e-05(6.7e-05)≈	8.02e-05(7.8e-05)	4.47e+14(6.2e+13)+	4.09e+14(5.4e+13)≈	4.15e+14(6.4e+13)
Adult	5.39e-05(3.3e-05)-	4.51e-05(3.3e-05)+	5.03e-05(3.1e-05)	5.78e+14(1.9e+13)+	5.04e+14(1.8e+13)-	5.34e+14(1.9e+13)
Drug consumption	1.74e-04(9.9e-05)-	1.69e-04(1.1e-04)≈	1.54e-04(8.5e-05)	4.08e+14(4.4e+13)+	3.90e+14(5.0e+13)+	3.76e+14(3.8e+13)
Patient treatment	4.24e-04(2.0e-04)-	2.54e-04(8.1e-05)≈	2.77e-04(1.2e-04)	4.60e+14(1.3e+14)+	3.51e+14(9.3e+13)≈	3.51e+14(1.1e+14)
LSAT	2.23e-04(8.2e-05)-	1.49e-04(5.2e-05)+	1.80e-04(7.2e-05)	2.98e+14(6.0e+13)+	2.34e+14(6.4e+13)≈	2.36e+14(6.7e+13)
Default	6.93e-05(3.1e-05)-	5.30e-05(2.6e-05)+	5.78e-05(2.9e-05)	3.46e+14(2.0e+13)+	3.07e+14(1.8e+13)≈	3.04e+14(1.7e+13)
Dutch	5.22e-05(1.3e-05)-	2.76e-05(1.2e-05)+	3.85e-05(1.4e-05)	6.14e+14(2.1e+13)+	5.55e+14(2.0e+13)-	5.76e+14(2.3e+13)
+/≈/-	0/1/11	7/4/0	-	12/0/0	1/6/5	-

Dataset	SP			HV		
	MOEL	$MOEL_{Rep}$	*FaMOEL*	*MOEL*	$MOEL_{Rep}$	*FaMOEL*
Heart health	0.661(9.1e-02)≈	0.678(0.15)-	0.660(9.8e-02)	14.7(8.5)≈	17.4(9.4)+	15.1(8.3)
Titanic	0.590(5.2e-02)≈	0.600(6.5e-02)≈	0.597(5.9e-02)	0.586(0.35)≈	0.621(0.36)+	0.578(0.37)
German	0.718(0.13)≈	0.700(0.12)≈	0.719(0.14)	43.6(30)-	45.4(31)≈	44.2(30)
Student performance	0.706(0.13)≈	0.679(0.12)≈	0.702(0.13)	36.8(21)-	41.9(22)≈	42.0(23)
COMPAS	0.529(7.3e-02)≈	0.532(7.2e-02)≈	0.526(8.0e-02)	20.7(8.6)≈	22.2(9.9)≈	21.4(8.6)
Bank	0.528(6.6e-02)≈	0.481(7.8e-02)+	0.528(7.4e-02)	17.4(5.0)-	19.0(4.9)-	20.5(5.8)
Adult	0.492(5.3e-02)≈	0.458(5.3e-02)+	0.500(5.2e-02)	4.58(0.79)-	4.99(0.70)-	5.48(1.1)
Drug consumption	0.485(5.7e-02)≈	0.455(5.8e-02)+	0.486(6.1e-02)	12.3(5.7)-	12.2(5.7)-	12.9(5.7)
Patient treatment	0.709(0.13)≈	0.639(0.15)≈	0.663(0.17)	42.3(14)-	45.9(15)≈	47.3(15)
LSAT	0.439(5.1e-02)≈	0.383(0.11)+	0.456(7.7e-02)	67.0(9.59)-	72.9(9.3)-	75.6(11)
Default	0.408(5.3e-02)≈	0.369(5.5e-02)+	0.412(5.1e-02)	21.2(2.34)-	24.6(2.72)-	25.7(2.5)
Dutch	0.523(5.8e-02)-	0.483(5.8e-02)≈	0.463(5.3e-02)	3.75(0.51)-	4.17(0.46)-	4.37(0.66)
+/≈/-	0/11/1	5/6/1	-	0/3/9	2/4/6	-

Student performance, COMPAS, Bank, Adult, Patient treatment, Default, and Dutch. Additionally, $MOEL_{Rep}$ performs better than *MOEL* on these datasets. The observation suggests that the subset can represent the entire objectives to some extent. However, this subset used in $MOEL_{Rep}$ may not serve as an optimal representative subset to guide the model training when compared to our proposed *FaMOEL*. In contrast, *FaMOEL* leverages fairness awareness based on the current evolution process to select a more suitable subset representing the 26 objectives. Therefore, the adaptively determined subset instead of a predefined static one is more suitably used to guide the model training.

Additionally, the distribution properties of the final solution set, including convergence, spread, and uniformity are used in further analysis as summarized in Table VI. In terms of the overall performance measured by HV, *FaMOEL* outperforms both $MOEL_{Rep}$ and *MOEL*. The pairwise win/tie/loss counts of $MOEL_{Rep}$ and *MOEL* against *FaMOEL* are 2/4/6 and 0/3/9, respectively.

Since, *MOEL* optimises all the 26 objectives, the effectiveness of *MOEL* may degrade, leading to worse GD but better PD in comparison with *FaMOEL*. This can be attributed to the fact that a solution set that is far away from the Pareto front may exhibit worse convergence but better spread. However, for the model set obtained by *MOEL*, a significant loss in convergence (GD) leads to worse overall performance (HV). On the contrary, $MOEL_{Rep}$ demonstrates better GD but worse PD, indicating that the model set obtained by $MOEL_{Rep}$ converges to a subregion of the Pareto front. That is, because $MOEL_{Rep}$ only optimizes the static subset and does not care about other considered objectives, which may make the models trap in the local regions considering all the objectives. What is more, on the seven datasets that are utilized to find a representative subset in $MOEL_{Rep}$, our proposed *FaMOEL* performs no worse than $MOEL_{Rep}$ on 5 out of 7 datasets without any prior knowledge. Considering all the remaining datasets, *FaMOEL* achieves better performance than $MOEL_{Rep}$ on 4 out of 5 datasets and no worse than $MOEL_{Rep}$ on the other dataset. This observation further verifies the effectiveness of our fairness-aware strategy using MOEL. More appropriate representative objectives can be constructed by *FaMOEL* to guide the model training process.

In Titanic and Heart health datasets, *FaMOEL* has worse HV performance compared to $MOEL_{Rep}$. In Titanic, although *FaMOEL* and $MOEL_{Rep}$ have the same performance in terms of GD, PD, and SP, *FaMOEL* has a slightly worse HV performance. One possible reason is that the HV may have a bias when measuring the overall performance [52] between the solution sets with very close convergence, spread, and uniformity. For Heart health, $MOEL_{Rep}$ demonstrates better convergence and further achieves better overall performance, albeit with slightly worse PD and SP.

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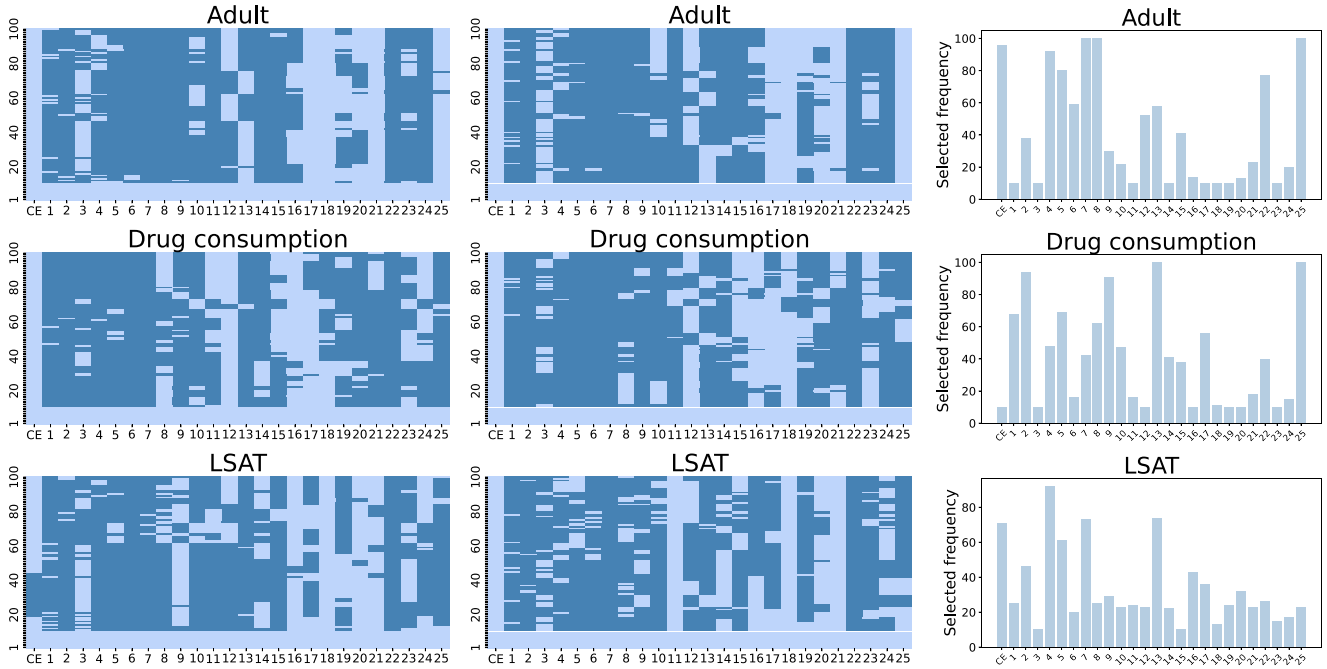


Fig. 3. Visualization of the fairness awareness process of *FaMOEL*. The first and second columns depict the evolution of the representative objective subset to be optimized at each generation. Each light-colored block represents the selection of an objective (corresponding to its respective column) for optimization at the corresponding generation (corresponding to the row). The third column displays the average frequency of selecting each objective along with 100 generations over 50 trials.

Then, we take a closer look at the fairness awareness process of our proposed *FaMOEL*. We present the results for Adult, Drug consumption, and LSAT datasets as examples, but the conclusions drawn from these datasets can be generalized to the remaining datasets as well. The visualization of the fairness awareness process is depicted in Fig. 3. The first and second columns in Fig. 3 illustrate the representative subset selected as optimization objectives at each generation, where a light-colored block indicates that the objective (associated with its respective row) is selected for optimization in the corresponding generation (determined by its column). Two arbitrary trials are plotted for demonstration purposes. Additionally, the averaged frequency of objective selection over 50 trials is presented in the final column of Fig. 3, providing an overview of the selection patterns across the entire trial.

Fig. 4 illustrates the correlation among accuracy and 25 fairness measures at generations 1, 50, and 100 in a single trial when dealing with the dataset Drug consumption. A correlation value close to 1 indicates a stronger positive correlation, while a value close to -1 suggests a stronger negative correlation. The findings reveal dynamic changes in the correlations among the measures, highlighted by the red boxes, throughout the training process. For example, the correlations between Fair 4 and Fair 16–Fair 24 vary across generations: they are positively correlated at generation 1 but negatively correlated at generation 50. Furthermore, we illustrate the size of the selected representative subset (i.e., number of objectives) across generations on the LSAT dataset in Fig. 5, demonstrating that the number of objectives varies from generation to generation.

Figs. 3–5 highlight the capability of our framework to identify a better representative subset along with the model

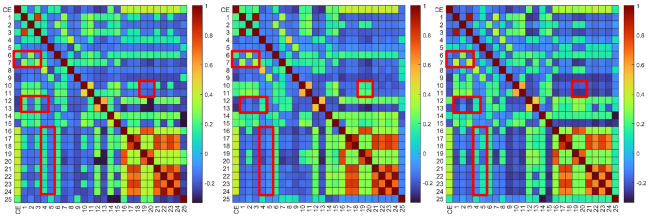


Fig. 4. Heatmap illustrating the correlation among accuracy and 25 fairness measures at generations 1, 50, and 100, respectively, in dealing with Drug consumption.

training process. These processes have three aspects: 1) across different model training stages within the same trial; 2) across different trials conducted on the same dataset; and 3) across different datasets. The adaptively determined representative subset obtained by our framework according to current training stages is more suitable as optimization objectives. Based on the experimental results, we conclude for these three cases as follows. In cases 1) and 2), optimizing a static subset may lead to the local regions and potentially result in worse performance. This can be observed by the stabilized curves of $MOEL_{\text{Rep}}$ in the later stages of evolution. It becomes crucial to identify a more suitable representative subset that can help overcome these local regions. The dynamic awareness of representative objectives in our proposed framework enables the exploration of different subsets, ultimately leading to improved performance. Regarding case 3), as demonstrated by previous research [6], the relationships among fairness objectives can vary depending on the dataset characteristics. Consequently, there is no universally applicable "perfect" subset that can adequately and comprehensively represent all

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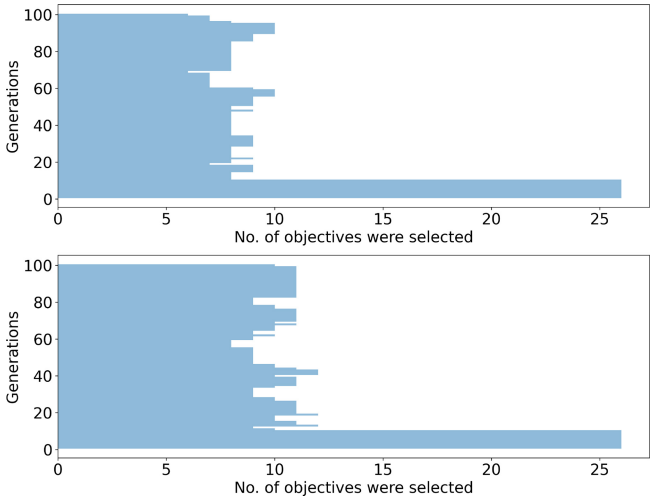


Fig. 5. Number of measures selected as objectives at each generation within two arbitrary trials on LSAT dataset.

the objectives across all the datasets. It further emphasizes the need for our fairness awareness approach that can tailor the representative subset to the specific dataset being considered.

FaMOEL using the fairness-aware strategy can adaptively select a properly representative subset of objectives according to the current process to guide the evolution of the population, which also does not rely on any prior knowledge.

D. Comparison With Optimising Frequently Selected Objectives

As depicted in the final column of Fig. 3, the selection frequencies of different objectives among *CE* and f_1–f_{25} vary across different datasets. It is intriguing to analyse the comparison between *FaMOEL* and the method optimizing only the frequently selected, denoted as $\text{MOEL}^{-}_{\text{Rep}}$. In this study, we specifically compare the performance of *FaMOEL* with that of $\text{MOEL}^{-}_{\text{Rep}}$, where $\text{MOEL}^{-}_{\text{Rep}}$ optimizes different objectives while maintaining the same settings for the remaining factors as in MOEL_{Rep}. For each dataset, the objectives whose frequency is larger than 60 among the ten generations are selected as the optimized objective set in $\text{MOEL}^{-}_{\text{Rep}}$, which is summarized in Table VII. Table VIII presents the results of GD, PD, SP, and HV obtained by $\text{MOEL}^{-}_{\text{Rep}}$ and *FaMOEL*, respectively. The results indicate that on the 7 out of 12 datasets, *FaMOEL* outperforms $\text{MOEL}^{-}_{\text{Rep}}$, while on the 11 out of 12 datasets, *FaMOEL* achieves no worse performance than $\text{MOEL}^{-}_{\text{Rep}}$. Specifically, *FaMOEL* demonstrates better convergence and spread performance as observed from the GD and PD measures.

In the Dutch dataset, *FaMOEL* performs worse than $\text{MOEL}^{-}_{\text{Rep}}$ in terms of HV. As Table VIII shows, *FaMOEL* performs better in GD and PD and the same in SP. We further measure the distribution property of the extremeness of the model set. For a model in each trial, the minimal angle denoted as a between the objective values and each axis is viewed as the extremeness of the model. A model with a closer to 45° indicates less serious extremeness and is a more centered distribution. Then, in Fig. 6, we plot the frequency histogram

TABLE VII
FREQUENTLY SELECTED OBJECTIVES OF *FaMOEL*, WHICH ARE USED AS OPTIMIZED OBJECTIVES OF $\text{MOEL}^{-}_{\text{Rep}}$

Dataset	Objectives
Heart health	CE, f_9, f_{11}, f_{15}, f_{25}
Titanic	CE, f_8, f_{11}, f_{15}, f_{25}
German	f_4, f_5, f_8, f_{10}, f_{11}, f_{22}, f_{25}
Student performance	CE, f_9, f_{10}, f_{11}, f_{22}, f_{25}
COMPAS	CE, f_2, f_7, f_8, f_9, f_{10}, f_{25}
Bank	CE, f_2, f_7, f_8, f_9,f_{10}, f_{11}, f_{15}, f_{25}
Adult	CE, f_4, f_5, f_8, f_9, f_{22}, f_{25}
Drug consumption	CE, f_2, f_4, f_8, f_{10}, f_{11}, f_{25}
Patient treatment	CE, f_4, f_5, f_8, f_{22}, f_{25}
LSAT	f_2, f_4, f_5, f_8, f_9, f_{10}, f_{25}
Default	CE, f_1, f_4, f_5, f_6, f_7, f_8, f_9, f_{10}, f_{22}, f_{25}
Dutch	CE, f_2, f_6, f_8, f_9, f_{10}, f_{15}, f_{25}

of the angle a of each model in the final set obtained by $\text{MOEL}^{-}_{\text{Rep}}$ and *FaMOEL* over 50 trials, respectively. Fig. 6 shows that the model sets obtained by $\text{MOEL}^{-}_{\text{Rep}}$ (the black line) are more around the center area, which contributes to better HV performance [52].

In summary, the above observations validate the effectiveness of *FaMOEL*. The representative subset should be adaptively determined according to the model training stage.

E. Effectiveness of Our Fairness-Aware Strategy

Our fairness-aware enhancement strategy, as described in Section III-B4, is designed to improve the work [17]. In order to evaluate the effectiveness of our enhancement, we compare our approach *FaMOEL* with the method that utilizes the original strategy proposed [17], denoted as *FaMOEL*$^{-}$. The difference between *FaMOEL* and *FaMOEL*$^{-}$ was discussed in Section III-B4. *FaMOEL*$^{-}$ is implemented with the same settings as ones of *FaMOEL* except for the fairness-aware strategy.

Table IX presents the results of GD, PD, SP, and HV obtained by *FaMOEL*$^{-}$ and *FaMOEL*, respectively. *FaMOEL* outperforms *FaMOEL*$^{-}$ and achieves no worse performance on the 10 out of 12 datasets. *FaMOEL*$^{-}$ demonstrates better convergence and uniformity performance as measured by GD and SP, respectively. However, *FaMOEL*$^{-}$ exhibits weaker spread performance. This suggests that the model set obtained by *FaMOEL*$^{-}$ only converges to a subregion, resulting in a significant loss in overall performance, despite being closer to the Pareto front compared to *FaMOEL*. Thus, our enhancement strategies are specifically designed to prevent the model set from getting trapped in local regions, such as designing a warm starting and constructing a robust mNCIE matrix. Also, the selection threshold τ is to avoid deleting objectives that are weakly positively correlated with $\mathcal{E}_J$ (line 17 in Algorithm 2). This strategy ensures that $\mathcal{E}'$ can adequately represent the entire $\mathcal{E}$.

In Student performance and Dutch, *FaMOEL* performs worse than *FaMOEL*$^{-}$ in terms of HV. In Student performance, compared with *FaMOEL*$^{-}$, *FaMOEL* exhibits better PD performance but worse GD and SP performance. This suggests that the model set obtained by *FaMOEL* is situated too far away from the Pareto front and loses a significant overall

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TABLE VIII
GD, PD, SP, AND HV VALUES OF FINAL MODEL SET AVERAGED OVER 50 TRIALS. "+/≈/−" INDICATES THAT THE AVERAGE INDICATOR VALUE OF THE CORRESPONDING ALGORITHM (SPECIFIED BY COLUMN HEADER) IS STATISTICALLY BETTER/SIMILAR/WORSE THAN THE ONE OF *FaMOEL* ACCORDING TO THE FRIEDMAN TEST WITH A 0.05 SIGNIFICANCE LEVEL. THE BEST AVERAGED VALUES ARE HIGHLIGHTED IN GRAY

Dataset	GD		PD		SP		HV	
	$MOEL^{-}_{Rep}$	$FaMOEL$	$MOEL^{-}_{Rep}$	$FaMOEL$	$MOEL^{-}_{Rep}$	$FaMOEL$	$MOEL^{-}_{Rep}$	$FaMOEL$
Heart health	2.66e-04(9.3e-05)+	2.88e-04(8.4e-05)	1.42e+14(3.1e+13)-	2.08e+14(4.5e+13)	0.710(9.6e-02)-	0.660(9.8e-02)	10.0(7.7)-	15.1(8.3)
Titanic	1.39e-04(7.8e-05)-	1.25e-04(8.9e-05)	4.04e+14(5.1e+13)-	4.40e+14(4.0e+13)	0.609(6.5e-02)≈	0.597(5.9e-02)	0.547(0.38)≈	0.578(0.37)
German	5.74e-04(2.2e-04)-	5.24e-04(2.1e-04)	4.05e+14(1.0e+14)≈	3.92e+14(1.1e+14)	0.700(0.11)≈	0.719(0.14)	39.4(29)-	44.2(30)
Student performance	4.66e-04(1.0e-04)+	5.04e-04(1.6e-04)	3.78e+14(5.1e+13)-	4.06e+14(9.3e+13)	0.656(0.11)+	0.702(0.13)	39.5(19)≈	42.0(23)
COMPAS	1.90e-04(5.0e-05)≈	2.00e-04(6.6e-05)	3.76e+14(4.8e+13)-	4.07e+14(4.6e+13)	0.569(7.9e-02)-	0.526(8.0e-02)	22.1(9.0)≈	21.4(8.6)
Bank	1.03e-04(6.6e-05)-	8.02e-05(7.9e-05)	4.2e+14(5.4e+13)+	4.15e+14(6.4e+13)	0.500(7.2e-02)+	0.528(7.4e-02)	19.6(5.2)-	20.5(5.8)
Adult	7.30e-05(1.3e-05)-	5.03e-05(3.1e-05)	4.9e+14(1.9e+13)-	5.34e+14(1.9e+13)	0.574(5.0e-02)-	0.500(5.2e-02)	5.66(1.0)≈	5.48(1.1)
Drug consumption	2.13e-04(1.1e-04)-	1.54e-04(8.6e-05)	2.99e+14(3.5e+13)-	3.76e+14(3.8e+13)	0.485(6.8e-02)≈	0.486(6.1e-02)	7.45(4.7)-	12.9(5.7)
Patient treatment	2.68e-04(7.8e-05)+	2.77e-04(1.2e-04)	2.83e+14(8.9e+13)-	3.51e+14(1.1e+14)	0.640(0.10)≈	0.663(0.17)	40.8(20)-	47.3(15)
LSAT	2.32e-04(1.0e-04)-	1.80e-04(7.3e-05)	2.30e+14(4.2e+13)≈	2.36e+14(6.7e+13)	0.412(8.6e-02)+	0.456(7.7e-02)	61.6(11)-	75.6(11)
Default	7.22e-05(2.4e-05)-	5.78e-05(3.0e-05)	3.03e+14(1.8e+13)≈	3.04e+14(1.7e+13)	0.411(5.5e-02)≈	0.412(5.1e-02)	24.9(2.7)-	25.7(2.5)
Dutch	6.53e-05(6.0e-06)-	3.85e-05(1.4e-05)	5.33e+14(2.1e+13)-	5.76e+14(2.3e+13)	0.484(3.8e-02)≈	0.463(5.3e-02)	4.82(0.55)+	4.37(0.66)
+/≈/-	3/1/8	-	1/3/8	-	3/6/3	-	1/4/7	-

TABLE IX
GD, PD, SP, AND HV VALUES OF FINAL MODEL SET AVERAGED OVER 50 TRIALS. "+/≈/−" INDICATES THAT THE AVERAGE INDICATOR VALUE OF THE CORRESPONDING ALGORITHM (SPECIFIED BY COLUMN HEADER) IS STATISTICALLY BETTER/SIMILAR/WORSE THAN THE ONE OF *FaMOEL* ACCORDING TO THE FRIEDMAN TEST WITH A 0.05 SIGNIFICANCE LEVEL. THE BEST AVERAGED VALUES ARE HIGHLIGHTED IN GRAY

Dataset	GD		PD		SP		HV	
	$FaMOEL^{-}$	$FaMOEL$	$FaMOEL^{-}$	$FaMOEL$	$FaMOEL^{-}$	$FaMOEL$	$FaMOEL^{-}$	$FaMOEL$
Heart health	2.60e-(1.0e-04)+	2.88e-(8.3e-05)	8.54e+(4.7e+13)-	2.08e+(4.5e+13)	0.537(0.11)+	0.660(9.8e-02)	10.6(7.7)-	15.1(8.3)
Titanic	1.32e-(7.8e-05)≈	1.25e-(8.8e-05)	2.47e+(1.1e+14)-	4.40e+(4.0e+13)	0.534(0.14)+	0.597(5.9e-02)	0.316(0.21)-	0.578(0.37)
German	3.95e-(1.7e-04)+	5.24e-(2.1e-04)	2.55e+(1.8e+14)-	3.92e+(1.1e+14)	0.632(0.25)+	0.719(0.14)	42.0(30)≈	44.2(30)
Student performance	4.10e-(1.0e-04)+	5.04e-(1.5e-04)	3.14e+(8.5e+13)-	4.06e+(9.3e+13)	0.614(0.12)+	0.702(0.13)	46.1(22)+	42.0(23)
COMPAS	1.48e-(5.6e-05)+	2.00e-(6.6e-05)	2.45e+(9.1e+13)-	4.07e+(4.6e+13)	0.390(0.11)+	0.526(8.0e-02)	20.9(9.4)≈	21.4(8.6)
Bank	8.88e-(5.5e-05)-	8.02e-(7.8e-05)	3.35e+(5.0e+13)-	4.15e+(6.4e+13)	0.438(6.7e-02)+	0.528(7.4e-02)	20.4(5.9)≈	20.5(5.8)
Adult	6.89e-(1.4e-05)-	5.03e-(3.1e-05)	4.89e+(3.9e+13)-	5.34e+(1.9e+13)	0.435(5.0e-02)+	0.500(5.2e-02)	4.66(1.2)-	5.48(1.1)
Drug consumption	1.35e-(5.4e-05)+	1.54e-(8.5e-05)	2.09e+(9.8e+13)-	3.76e+(3.8e+13)	0.369(0.13)+	0.486(6.1e-02)	11.3(6.0)-	12.9(5.7)
Patient treatment	1.59e-(7.9e-05)+	2.77e-(1.2e-04)	1.70e+(9.8e+13)-	3.51e+(1.1e+14)	0.345(0.19)+	0.663(0.17)	42.1(20)-	47.3(15)
LSAT	1.14e-(3.1e-05)+	1.80e-(7.2e-05)	1.37e+(6.6e+13)-	2.36e+(6.7e+13)	0.333(0.11)+	0.456(7.7e-02)	75.3(12)≈	75.6(11)
Default	5.28e-(1.8e-05)≈	5.78e-(2.9e-05)	2.37e+(3.3e+13)-	3.04e+(1.7e+13)	0.357(6.9e-02)+	0.412(5.1e-02)	25.3(2.7)≈	25.7(2.5)
Dutch	5.45e-(5.1e-06)-	3.85e-(1.4e-05)	4.19e+(4.4e+13)-	5.76e+(2.3e+13)	0.369(5.1e-02)+	0.463(5.3e-02)	4.88(1.1)+	4.37(0.66)
+/≈/-	7/2/3	-	0/0/12	-	12/0/0	-	2/5/5	-

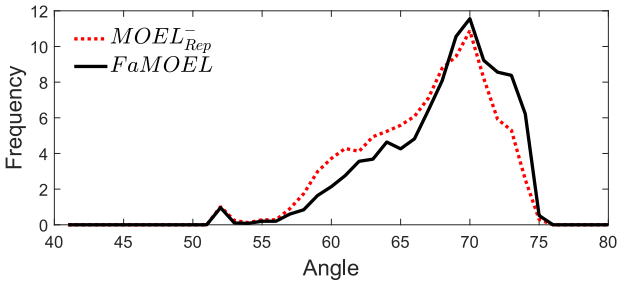


Fig. 6. Averaged frequency of minimal angles between each axis and each model of the final model set in the Dutch dataset over 50 trials.

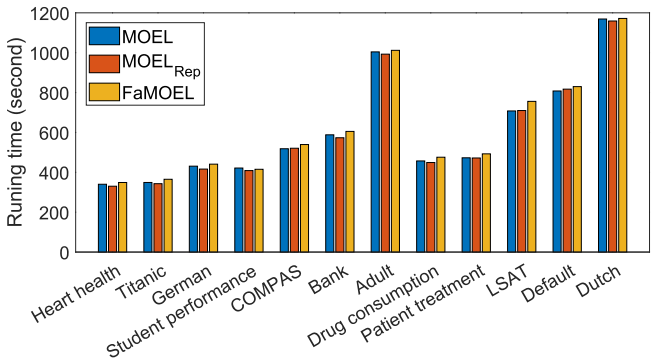


Fig. 7. Computational time of *MOEL*, MOEL$_{\text{Rep}}$, and *FaMOEL* on 12 datasets.

performance. As for Dutch, *FaMOEL* has better GD and PD but worse SP performance. One potential explanation for the inferior HV performance of *FaMOEL* is that the model set obtained by *FaMOEL* has poorer uniformity, which ultimately leads to a substantial degradation in overall performance.

In summary, our proposed fairness-aware enhancement strategy can contribute to constructing a more suitable representative subset of all the considered objectives to guide the model training process, especially in improving the spread performance of the learning models.

F. Computational Cost Analysis

To analyse the efficiency of *FaMOEL*, we report the average runtime of *MOEL*, MOEL$_{\text{Rep}}$, and *FaMOEL* in Fig. 7. Overall, *FaMOEL* have a similar computation runtime to *MOEL* and MOEL$_{\text{Rep}}$ as indicated in Fig. 7. Nonetheless, it is worth noting that MOEL$_{\text{Rep}}$ which involves two issues relies on a set of predefined fairness measures. First, determining suitable measures requires considerable computational cost as different

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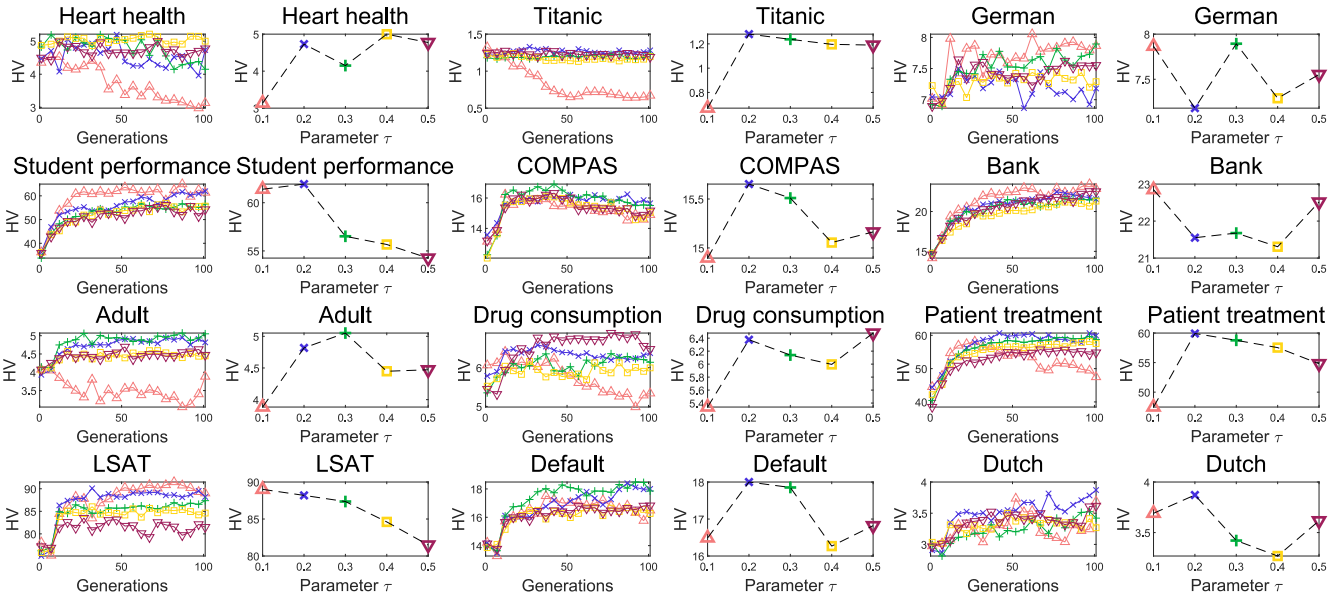


Fig. 8. Averaged HV values of *FaMOEL* under coarse-grained τ values $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. For each dataset, the left figure plots the averaged HV values along with generations. The right figure plots the averaged HV values in the final generation along with different parameters τ.

algorithms should be run across various datasets to identify the correlation among the measures. Second, those predefined measures may show different correlation on a new dataset. Therefore, the results demonstrate the effectiveness of our framework.

G. Parameter Sensitivity Analysis

In this study, we aim to analyse the sensitivity of the unique hyperparameter in our algorithms τ and recommend a value. The parameter τ is introduced in our enhanced fairness-aware strategy and serves as a selection threshold for determining the objective set *Del* to be removed. Each objective in *Del* is viewed to be highly positively correlated with the most representative objective $\mathcal{E}_J$ in the objective set indexed by S (cf. Algorithm 2). The sensitivity analysis of τ involves two steps: 1) applying a set of coarse-grained τ values $\{0.1, 0.2, 0.3, 0.4, 0.5\}$ to all the 12 datasets and 2) selecting two datasets that exhibit higher sensitivity to τ and conducting a fine grained analysis using a set of τ values to these two datasets.

Fig. 8 presents how the HV performance of *FaMOEL* varies with the τ set $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. The influences of τ exhibit diverse patterns across the 12 datasets but have a similar observation. Considering all the 12 datasets, except for Heart health and Drug consumption, the better performance of HV falls into the interval $[0.1, 0.3]$. In addition, it also demonstrates that both Patient treatment and LSAT show a higher sensitivity to τ. To further investigate the influence of τ in a more detailed manner, we conduct a fine-grained analysis using a set of τ values ranging from 0.1 to 0.3 with intervals of 0.02 as depicted in Fig. 9. Based on the results on these two datasets, a selection threshold value of 0.22 appears to be a preferable choice for τ. In general, τ is a problem dependent hyperparameter.

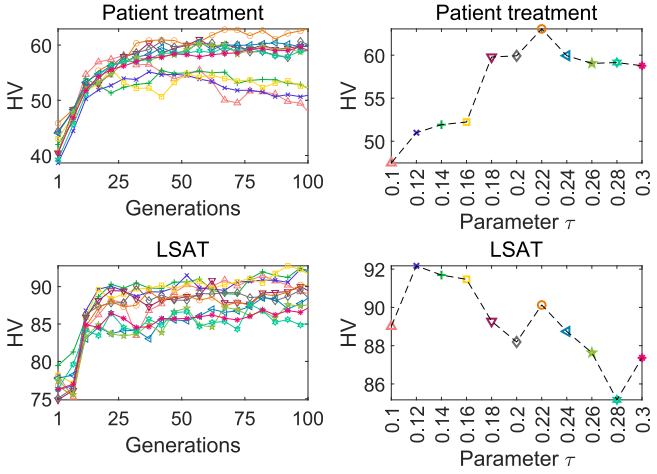


Fig. 9. Averaged HV values of *FaMOEL* under fine-grained τ values ranging from 0.1 to 0.3 with intervals of 0.02. For each dataset, the left figure plots the averaged HV values along with generations. The right figure plots the averaged HV values in the final generation along with different parameters τ.

V. Conclusion

When considering a set of fairness measures, this article proposes to dynamically and adaptively determine a representative subset of measures as optimization objectives during the model training without relying on any prior knowledge. The determined set can be used as objectives of MOEL to guide the evolution of the learning models. Extensive experimental studies demonstrate that our framework achieves very good performance in dealing with accuracy and 25 fairness measures. Furthermore, it is observed that our fairness-aware strategy effectively detects and selects suitable objectives, which vary across different training stages. Compared with the state-of-the-art algorithm optimizing a static representative subset, our method eliminates the need for prior knowledge in determining the representative subset and achieves superior

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performance in general. It is also worth noting that our work represents one of the few attempts in machine learning where the learning objectives (or loss functions) change adaptively during training.

In the future work, we aim to address the two key aspects to improve our model's performance and applicability. As shown in Fig. 8, the optimal parameter τ varies across different datasets. Therefore, an adaptive mechanism for tuning the parameter τ is required to determine a more appropriate subset of representative measures along with the model training. We can also employ one of the existing methods for tuning parameters automatically [54]. Furthermore, we plan to enhance the feasibility and interoperability of our framework when applied to more complex models, such as the deep learning models.

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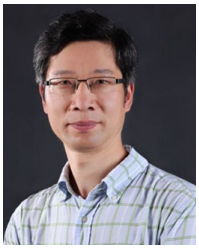
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